

Mathematical Framework for Individual Profile Vectors

Comparing Cosine Distance, Jaccard Similarity, and Spearman/Pearson Correlations

When designing vectors to model individual profiles across heterogeneous dimensions—specifically **Background, Interests, and Skills**—the selected mathematical metric dictates the fundamental philosophy of how similarities are calculated. This document evaluates four primary metrics, detailing their mechanics, unique advantages, and critical blind spots within this deployment architectural context.

Core Vector Comparison Metrics

Metric	Ideal Data Type	Profiling Philosophy	Critical Blindspot
Cosine Distance	Continuous, weighted numerical data (e.g., hours dedicated, years of experience).	Measures the angle between vectors. Focuses purely on the <i>relative proportions</i> of traits rather than absolute magnitude.	Completely blind to scale; a master and an apprentice with identical interest ratios appear identical.
Jaccard Similarity	Binary, Boolean features or categorical tag checklists (e.g., has certification = 1, does not = 0).	Measures <i>strict overlap</i> between active sets. Calculated as intersection divided by union.	Fails to account for intensity, frequency, or proficiency levels of attributes.
Spearman Correlation	Ordinal values, ordered priorities, or subjective ratings (e.g., scale of 1-5).	Evaluates <i>relative prioritization</i> . Converts values to ordinal ranks before matching trend directions.	Unstable on highly sparse vectors where users share very few common overlapping dimensions.
Pearson Correlation	Linear numeric attributes; mathematically equivalent to mean-centered Cosine Similarity.	Evaluates strict <i>linear dependencies</i> between raw scores, detecting proportional escalations.	Highly sensitive to extreme outliers and vulnerable to subjective score inflation.

Architectural & Domain Implementation Challenges

1. The Heterogeneity Trap

Combining Background (categorical/nominal), Interests (ordinal/subjective), and Skills (continuous metric/proficiency) into a single vector forces a mathematical compromise. Jaccard naturally treats all items as discrete tags but discards skill levels. Cosine elegantly weights skills but distorts non-ordered background categorical indices unless they are heavily manipulated via one-hot encoding.

2. Correcting for User Rating Inflation (Spearman's Edge)

When individuals self-rate interests or skills on a scale (e.g., 1 to 5), they display cognitive biases. An optimistic user might rate most skills a 5, while a conservative user rates equal proficiency as a 3. Pearson

and Cosine penalize this variance as a distance mismatch. Spearman bypasses this completely by converting both profiles into internal ranks (1st priority, 2nd priority), matching them on their internal value hierarchy rather than absolute numbers.

3. Zero-Inflation and High Sparsity

In large-scale system catalogs, the universal pool of skills and interests may span thousands of dimensions, yet an individual might only possess 15. This results in highly sparse vectors dominated by zeros. While Cosine and Pearson calculate similarity across the entire empty space (unless explicitly filtered), Jaccard naturally ignores dimensions where both individuals score 0, prioritizing only the actively populated intersect.

Strategic Recommendation

To capture the best of all worlds without distorting the data, avoid a single massive composite vector. Instead, build a **multi-tier ensemble matching engine**: evaluate Background using Jaccard Similarity, Interests via Spearman Rank Correlation to neutralize scale bias, and Skills using Cosine Similarity to maintain strict proficiency weighting. Linearly combine these independent scores into a final weighted composite index.